



cardiac Dysrhythmia

Prepared by
Dr Ghada Elsayed



Dysrhythmia

Outlines

- **Definition of dysrhythmia**
- **Sinus dysrhythmias**
 - Sinus tachycardia:**
 - Sinus Bradycardia:**
- **Atrial dysrhythmias**
 - **Atrial Flutter**
 - Atrial Fibrillation**
- **Ventricular dysrhythmia**
 - ventricular tachycardia**
 - Ventricular fibrelation**



Pre test

Mention types of
dysrhythmia

Mention
management of
ventricular
fibrillation

Pre test

- Count the following rate







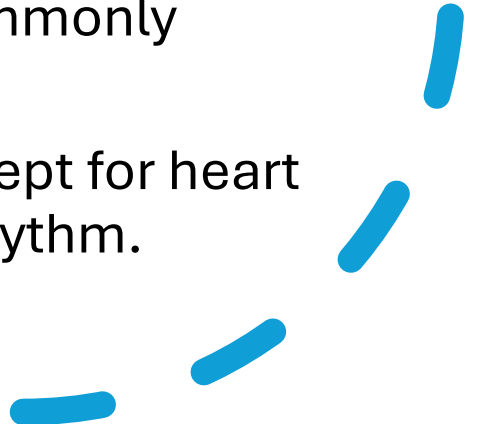
- Sinus Dysrhythmias
- 1-Sinus Tachycardia

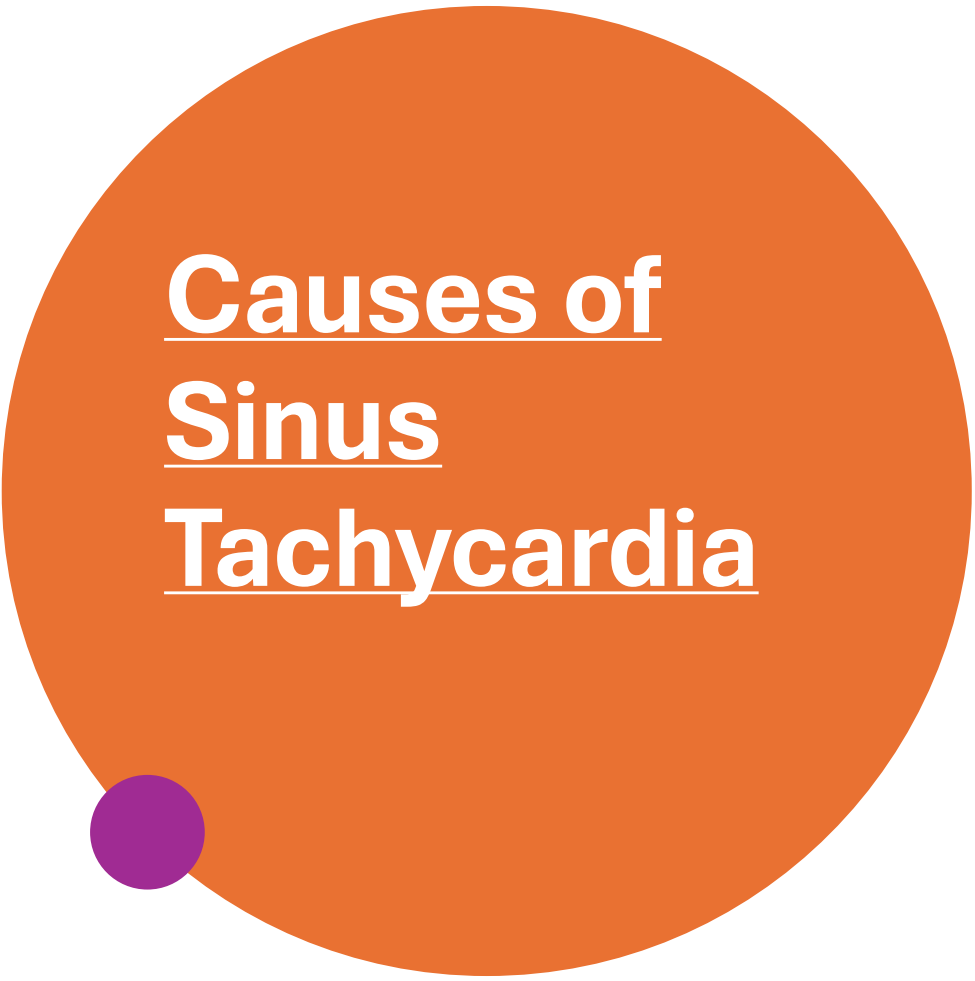
Sinus tachycardia

- **Sinus tachycardia** originates in the SA node & discharges impulses regularly at a rate between 100 – 160 times/minute.
- Sinus tachycardia meets all the criteria for normal sinus rhythm except that the rate is greater than 100 beat/min.
- It is a normal response to stimulation of the sympathetic nervous system, also is a normal findings in children under age of 6 years

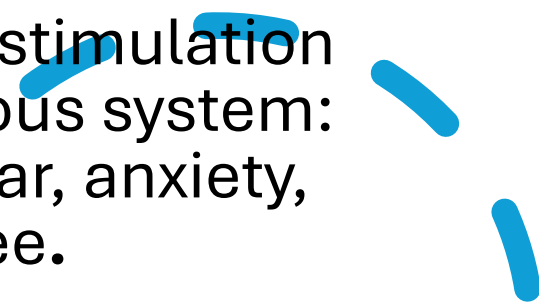


ECG Characteristics of Sinus Tachycardia

- Rhythm: Atrial and Ventricle Rhythm are regular
 - Rate: Atrial and Ventricle rate are greater than 100 beat/minute ,
 - P wave: normal size and configuration.
 - Precedes each QRS, duration from 0.06 to less than 0.12 second.
 - P Interval: within normal limit. (0.12 to 0.20 second)
 - QRS Complex: Normal duration and configuration.
 - Duration from 0.06 to 0.10 second in width.
 - T wave: Normal size and configuration.
 - QT Interval: within normal limit, but commonly shorted
 - (N.B) All other ECG characteristics, except for heart rate, are the same as in normal sinus rhythm.
- 



Causes of Sinus Tachycardia

- All normal responses to stimulation of the sympathetic nervous system: Exercise, pain, stress, fear, anxiety, exercise, excessive coffee.
 - *Cardiac condition: Heart failure, cardiogenic shock, pericarditis.
 - *Other condition: Hyperdynamic circulation, shock, anemia, respiratory distress, pulmonary embolism, hyperthyroidism.
 - *Drugs: Atropine and dopamine
 - *Alcohol
 - Nicotine
- 

Signs and Symptoms of Sinus Tachycardia

The patient will have a peripheral pulse rate above 100 beat/minute, but with regular rhythm (asymptomatic)



Sinus Tachycardia is detrimental in ischemic heart diseases patient because it decreases the time for ventricular filling, decreases stroke volume & thus compromises cardiac output.



Sinus tachycardia is often the earliest sign of Left Ventricular Failure (LVF), shock, or infarct extension.

Treatment of Sinus Tachycardia:

Treatment Goals

*Maintain adequate cardiac output
*Maintain adequate tissue perfusion

*Correct the underlying cause.

SINUS BRADYCARDIA



- **2-Sinus Bradycardia**
- Sinus bradycardia results when the SA node generates fewer than 60
- beat/minute

+

•

○

Causes of Sinus Bradycardia

- **Non cardiac causes**
- *Hyperkalemia, hypothermia, hypothyroidism
- *Increased intracranial pressure (BP increases & HR decreases and become irregular)
- **Cardiac causes**
 - Ischemia of the SA node (acute Inferior wall MI)
 - Cardiomyopath
 - **Drugs causes:**
- *Digoxin
- *Beta-blockers
 - Calcium channel blockers
- Finally, sinus bradycardia may occur normally in athletes persons

Signs and Symptoms of Sinus Bradycardia



*Asymptomatic bradycardia means no signs and symptoms of cardiac output appear



* Symptomatic bradycardia occurs when manifestations of low cardiac output appear



dizziness, blurred vision, altered mental status, chest pain, syncope, crackles,

Treatment of Sinus Bradycardia

Sinus bradycardia usually does not require treatment unless the patient displays symptoms of such as hypotension, dizziness, hypoperfusion, chest pain, or changes in level of consciousness. No treatment is required until symptom of low cardiac output appears. Correct the underline causes

-Administer drug such as:
Atropine, Dopamine, Epinephrine

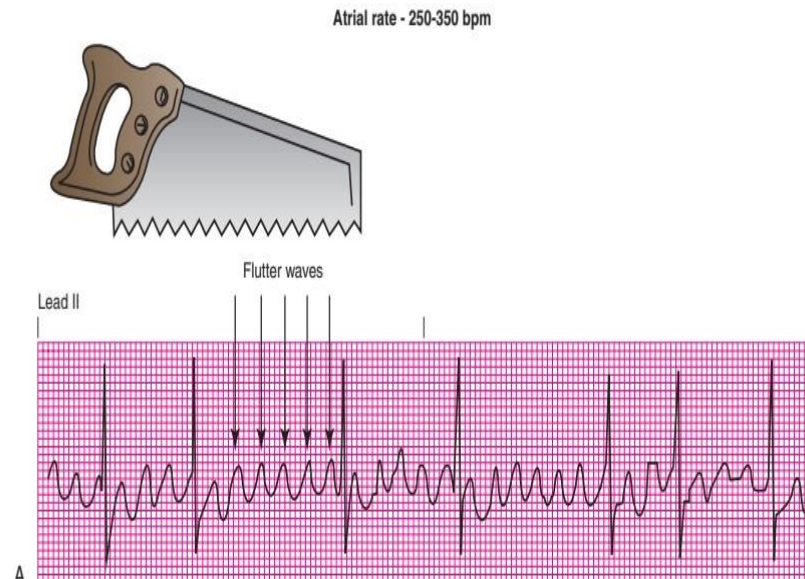
PARAMETERS	NORMAL SINUS RHYTHM	SINUS BRADYCARDIA	SINUS TACHYCARDIA
Rate	60-100/min	<60/min	>100/min
Rhythm	Regular	Regular	Regular
P wave	Present, with one per QRS	Present, with one per QRS	Present, with one per QRS
PR interval	0.12-0.20 sec and constant	0.12-0.20 sec and constant	0.12-0.20 sec and constant
QRS	0.06-0.10 sec	0.06-0.10 sec	0.06-0.10 sec

Comparison among normal sinus rhythm , sinus tachycardia & bradycardia

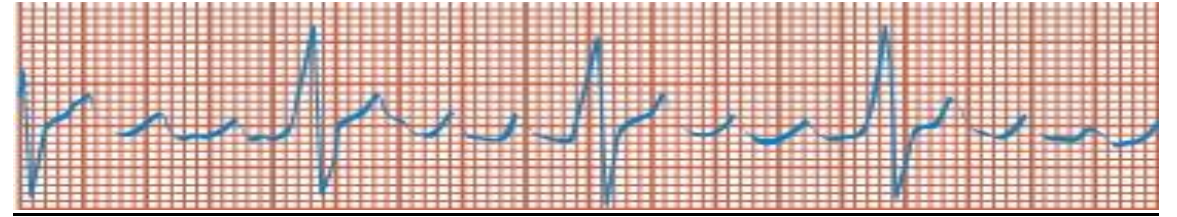


Atrial dysrthemia

Atrial Flutter



1-Atrial Flutter



Mechanism of Atrial flutter



- Atrial flutter is a condition in which there is an abnormal focus in the atrium that discharges rapid regular impulses but due to the protective mechanism in the AV node, which does not allow all of the atrial impulses to be conducted to the ventricles.

Characteristics of Atrial Flutter ECG Rhythm

Rhythm : Atrial rhythm is regular

Ventricular rhythm regular or may be irregular depend on the AV Rate; Atrial rate 250-350 b/min conduction pattern

P Wave: replaced by flutter (saw tooth)

QRS Complex: exhibits a normal configuration

PR interval: Cannot be measured.

P: QRS Ratio: Many: 1

Signs and symptoms of Atrial flutter

* Usually the patient is asymptomatic unless atrial flutter result in a tachycardia called Rapid Ventricular Response (RVR) that occurs when the ventricular response is greater than 100 beat/min.

* If Atrial Flutter occurs in conjunction with a rapid ventricular rate, the ventricular chambers cannot fill adequately, resulting in varying degrees of hemodynamic compromise

Causes of Atrial Flutter

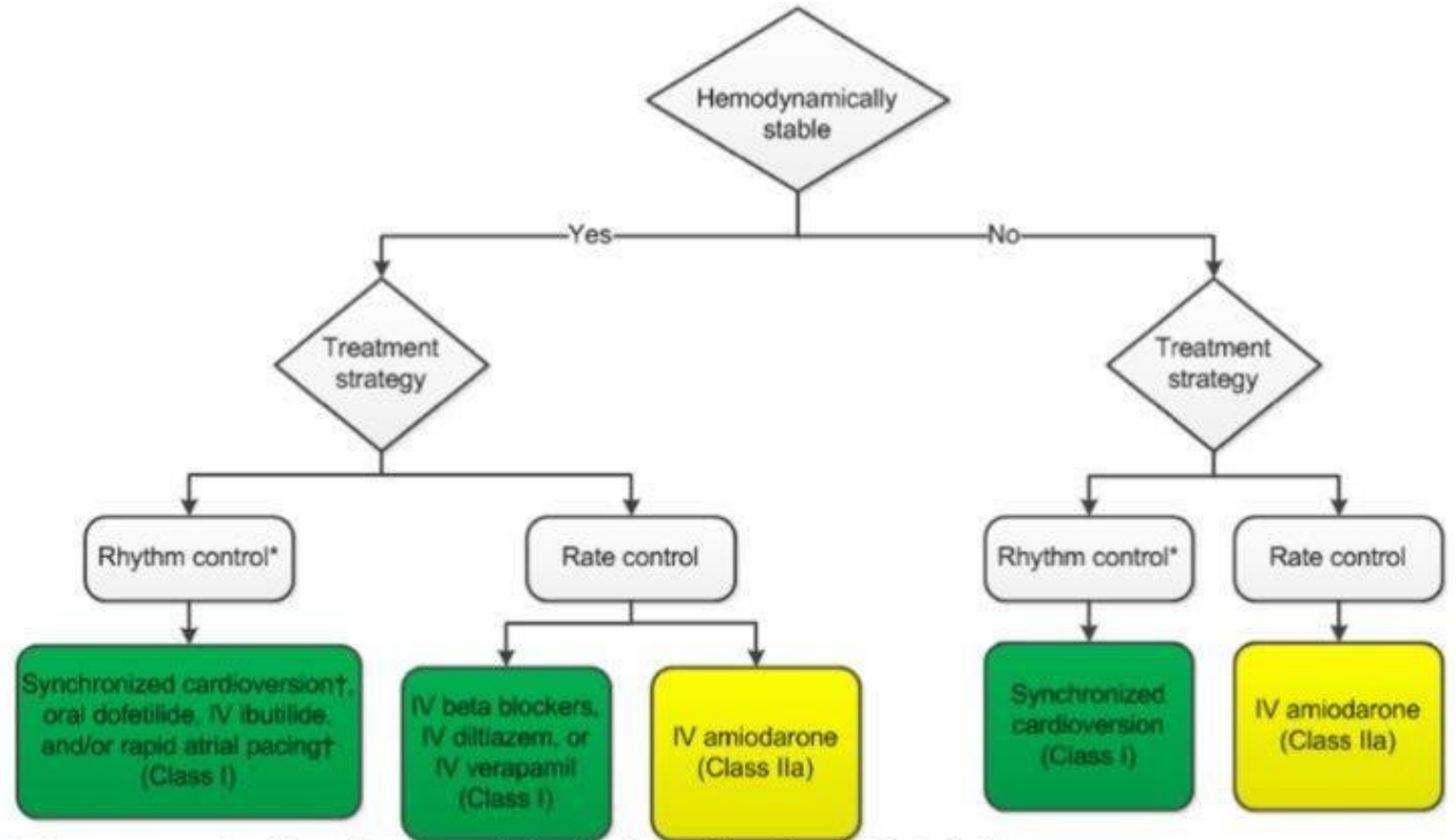
Atrial flutter often is seen in the presence of underlying cardiac disease, including

Coronary Artery Diseases (CAD),

Cor-pulmonale, and rheumatic heart disease.

Hypoxemia and alcoholism can cause Atrial Flutter.

• Treatment of Atrial Flutter



Colors correspond to Class of Recommendation in Table 1; drugs listed alphabetically.

*Anticoagulation as per guideline is mandatory.

†For rhythms that break or recur spontaneously, synchronized cardioversion or rapid atrial pacing is not appropriate.

IV indicates intravenous.

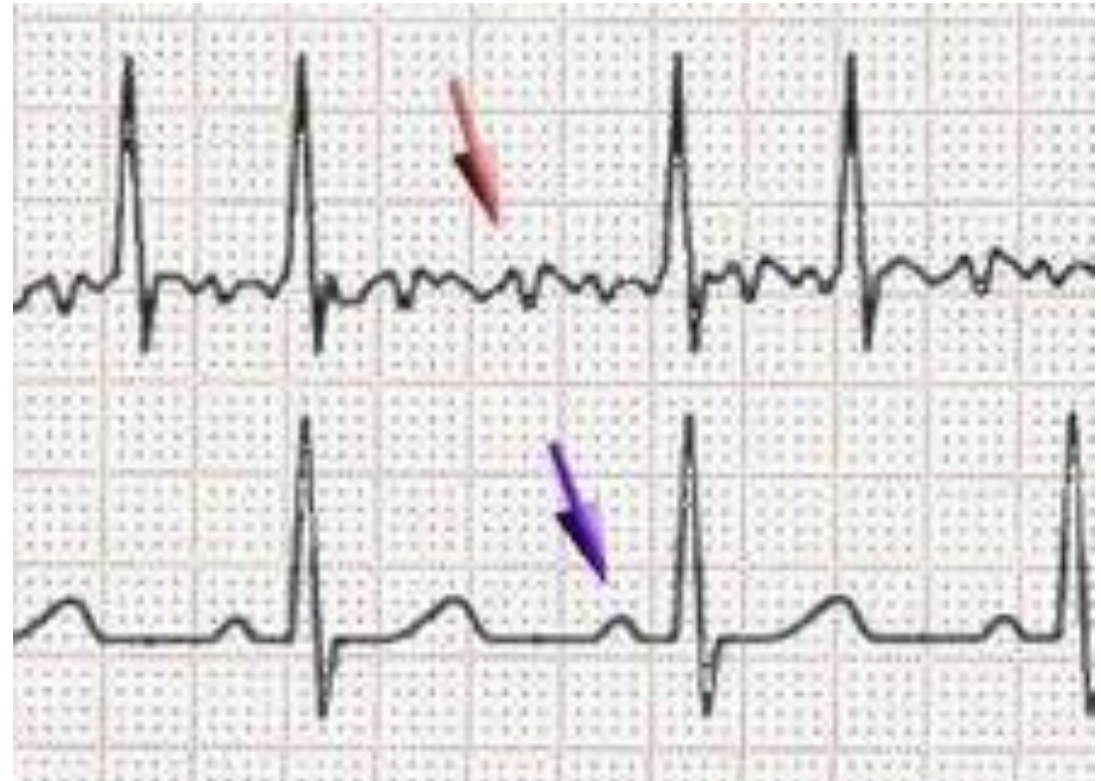
Figure 18 Acute treatment of atrial flutter.



- Antidysrhythmic Medications
- prescribed for two reasons:
 - *Convert the atrial flutter to sinus rhythm (rhythm control) and
 - *Slow conduction through the AV node (rate control).
- The most effective medications for rhythm control are Amiodarone, and Sotalol.

Atrial fibrillation

- **Definition:** Atrial fibrillation (AF or A-fib) is an abnormal heart rhythm characterized by rapid and irregular beating of the atria often it starts as brief periods of abnormal beating which become longer and possibly constant over time

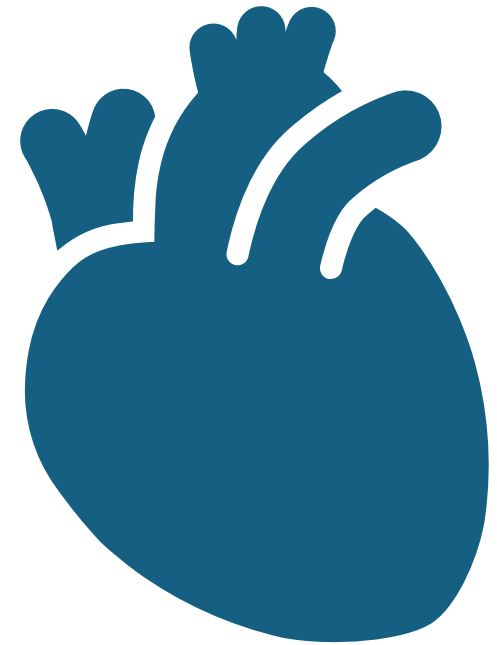




Heart Rate	Rhythm	P Wave	PR Interval (sec.)	QRS (Sec.)
Var.	Irregular	Wavy irregular	NA	<.12

Causes of Atrial fibrillation

- Cardiovascular factors known to be associated with the development of AF include:
- High blood pressure,
- Coronary artery disease,
- Heart valve disease
- Left atrial enlargement,
- hypertrophic cardiomyopathy (HCM),
- Pericarditis,



Signs and Symptoms of Atrial Fibrillation

Rapid and irregular heart rates may be perceived as

The sensation of the heart beating too fast, irregularly, or skipping beats (palpitations), tiredness, shortness of breath, dizziness or or exercise intolerance and feeling faint occasionally may produce anginal chest pain (if the high heart rate causes the heart's demand for oxygen to increase beyond the

Treatment of Atrial Fibrillation



The acute management of AF can be divided into 3 components:



Control of the heart rate,



Adenosine



Cardioversion (electrical and pharmacological),



Thromboprophylaxis.

Electrical Cardioversion

- More than 50% of episodes of recent-onset AF will spontaneously revert to sinus rhythm within the first 72 hours .
- For the remaining cases of AF that are complicated by hypotension, pulmonary edema, or myocardial ischemia, direct-current cardioversion is the appropriate intervention.
- An energy of 100 J is usually enough for successful cardioversion using biphasic shocks, but 200 J is recommended for the initial cardioversion attempt in the most recent guidelines on AF.



Pharmacological Cardioversion

- Drug-induced cardioversion is used in cases of uncomplicated AF that are refractory to rate control , or for first episodes of uncomplicated AF that are less than 48 hours in duration, to eliminate the need for anti coagulation .
- Several antiarrhythmic agents can be effective in terminating AF , **including amiodarone** .

Atrial Flutter vs. Atrial Fibrillation

Summary of Disease Characteristics

	Atrial Flutter	Atrial Fibrillation
Atrial Rate	250 to 400 bpm	400 bpm
Ventricular Rate/Rhythm	Usually regular	Varies with conduction Grossly irregular
Pattern	Saw tooth baseline	Irregular or almost flat baseline “Irregularly irregular”
Underlying Mechanism	Reentry via macro re-entrant circuit	Typically multiple wavelet reentry



Ventricular Dysrhythmias

- 1-Premature Ventricular Contractions (PVC)s
- An ectopic beat originating prematurely at the level of the ventricles.
- The beat is ventricular in origin and results in no electrical activity in the atria. As a result, no P waves appear.



1-Premature Ventricular Contractions (PVC)s

- Rhythm: Irregular
- Rate: Underlying rhythm
- P wave: None with PVC but P wave present with Other QRS complex.
- PR Interval: *Not measurable* QRS Complex: *Wide QRS*
- T wave: opposite direction from QRS
- Others: may be followed by compensatory pause

Causes of Premature Ventricular Contractions

- The most common of all ectopic beats, PVCs can occur with or without heart disease in any age group. They are especially common in people with myocardial disease (ischemia or infarction) or with myocardial irritability (hypokalemia, increased levels of catecholamines, or mechanical irritation with a wire or catheter).

Treatment of Premature Ventricular Contractions

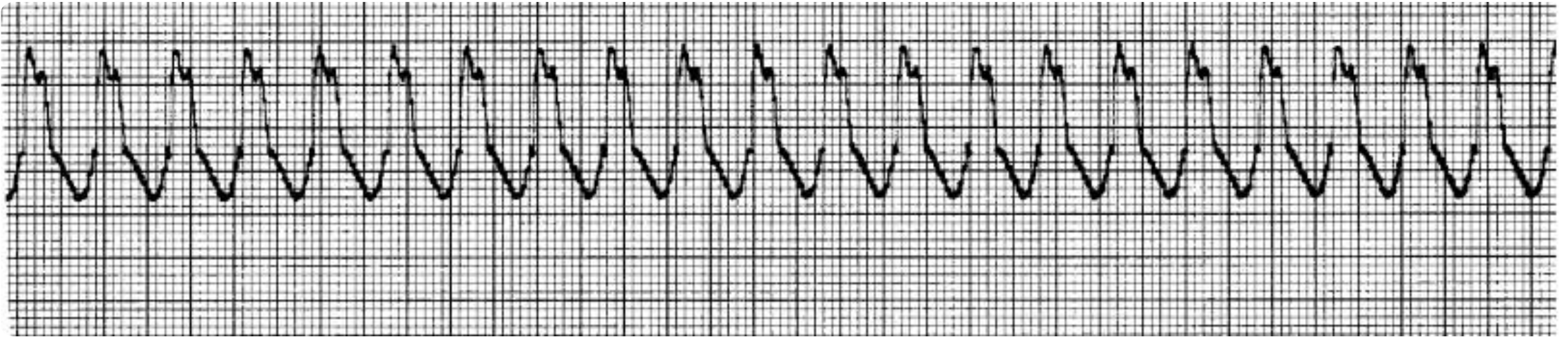
Infrequent, isolated PVCs require no treatment.

Multiple or consecutive PVCs may be managed with antiarrhythmic agents.

In the emergency setting, Amiodarone and Lidocaine are the drugs of choice.

If serum potassium is low, potassium replacement may correct the dysrhythmia.

If the dysrhythmia is caused by Digitalis toxicity, withdrawal of the Digitalis may correct it.



2-Ventricular Tachycardia (VT)



Ventricular Tachycardia



Characteristic of Ventricular Tachycardia (VT)



Rhythm: Ventricular rhythm regular



Rate: Rapid rate over 100 beat/min



QRS shape and duration: Greater than 0.12 seconds, bizarre, abnormal shape.(wide QRS complex)



P wave: absent



PR Interval: Not measurable



- T wave: opposite direction from QRS complex

Causes of Ventricular Tachycardia

Commonly occurs in myocardial ischemia or Infarction

Cardiomyopathy

Mitral Valve Prolapse

Congestive Heart Failure

Digitalis toxicity is a common cause of VT

Signs and Symptoms of Ventricular Tachycardia

- Signs and symptoms of hemodynamic compromise (e.g., ischemic chest pain, hypotension, pulmonary edema, and unconsciousness) may be seen if the rate is fast and the tachycardia is sustained.



Treatment of Ventricular Tachycardia

- If the patient is hemodynamically stable with the dysrhythmia, Lidocaine may be administered intravenously.
- If the patient becomes unstable, synchronized cardio version (or unsynchronized defibrillation in emergency situations) is indicated.
- Long-term treatment for this dysrhythmia may involve the use of an Implantable Cardioverter Defibrillator (ICD).

Anti dysrhythmia of VT

Lidocaine

- Treatment is IV administered Lidocaine (Xylocaine)
- The dose for ventricular tachycardia is 1 mg/kg to 1.5 mg/kg IV push.
-  with doses of 0.5 mg/kg to 0.75 mg/kg repeated every five to ten minutes
- Up to a maximum dose of 3 mg/kg.
- Following a successful lidocaine bolus, run a continuous IV infusion 2 Gm in 500 ml D5W.

Amiodarone

- * Patients with pulse less VT: New recommendation from the American Heart
- Association emphasize the initial use of 300 mg of IV administration Amiodarone, followed by repeated 150 mg IV doses every 8 to 10 minutes, in patients with pulse less VT.

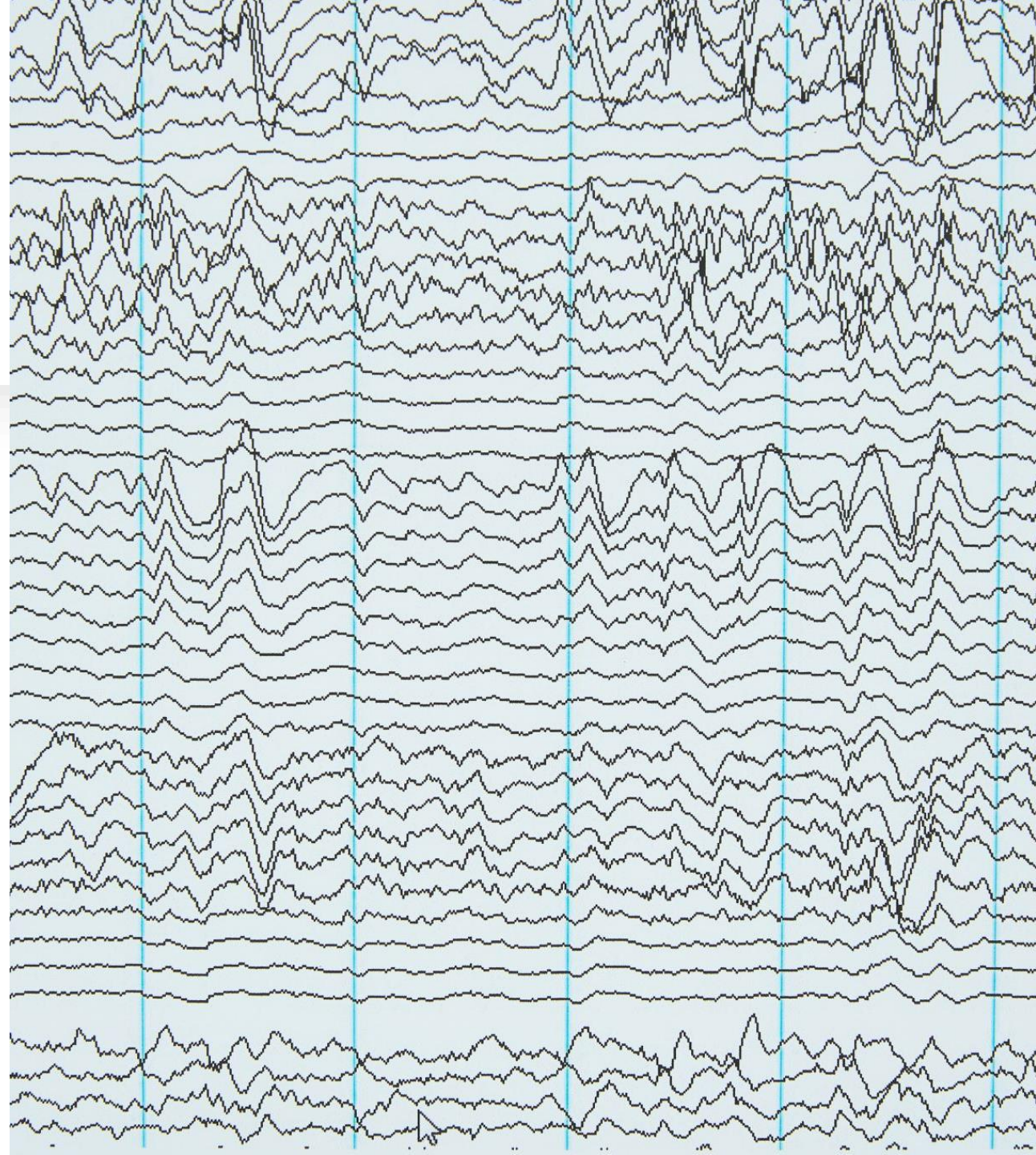
Ventricular Fibrillation

- **Definition:** Commonly called V-fib, is a chaotic pattern of electrical activity in the ventricles in which electrical impulses arise from many different foci. It produces no effective muscular contraction and no cardiac output.



ECG characteristics of Ventricular fibrillation

- Rate: Indeterminable
- Rhythm: Irregular, wavy baseline without recognizable QRS complexes
- P wave: Absent.
- Cannot be distinguished from fibrillating ventricular baseline.
- PR interval: Absent
- QRS duration: Absent.
- No QRS complexes present.
- QRS complex: Normal P wave and QRS are replaced by a wavy baseline.



Causes of Ventricular fibrillation

- Myocardial ischemia
- MI
- Untreated ventricular tachycardia
- Underlying heart disease
- Acid–base imbalance
- Electric shock
- Severe hypothermia
- Electrolyte imbalances, such as hypokalemia, hyperkalemia, and hypercalcemia
- Drug toxicity, including digoxin - severe hypoxia.

Treatment of Ventricular fibrillation



VF is a rapidly fatal rhythm, which virtually never terminates spontaneously.



CPR must be initiated promptly and rapid, asynchronous DCC performed as soon as possible.



A single shock of 200 to 360 J (biphasic devices, 200 J; monophasic devices, 360 J) should be given initially followed by immediate resumption of CPR for 2 minutes before checking for a pulse.



If VF/pulseless VT persists, an immediate second shock (biphasic devices, ≥ 200 J; monophasic devices, 360 J) should be given followed by a vasopressor (1 mg of epinephrine every 3 to 5 minutes; single dose of



40 units of vasopressin may replace first or second dose of epinephrine).



If VF/pulseless VT persists after two or three shocks, CPR, and a vasopressor, administration of an antiarrhythmic should be considered (amiodarone is preferred and lidocaine as an alternative).



Amiodarone Dose for Arrhythmias

- **Initial dose:** 1000 mg over the first 24 hours of therapy, delivered by the following infusion regimen:
- **Loading infusions:** 150 mg over the first 10 minutes (15 mg/min), followed by 360 mg over the next 6 hours (1 mg/min).
- **Maintenance infusion:** 540 mg over the remaining 18 hours (0.5 mg/min).
- **Maintenance dose:** After the first 24 hours, continue the maintenance infusion rate of 0.5 mg/min; may increase infusion rate to achieve effective arrhythmia suppression.
- **Supplemental infusions:** 150 mg over 10 minutes (15 mg/min) for breakthrough episodes of ventricular fibrillation (VF) or hemodynamically unstable ventricular tachycardia (VT)

Post Test

- Mention types of atrial dysrhythmia
- Mention management of ventricular tachycardia



Any question



Thank you